

Decision under Uncertainty and Portfolio Analysis

Lecture 4: IRFA and MMMEF

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September 21, 2006

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Early Models of Reference Dependence

- ▶ Markowitz (1952) first proposed defining utility in terms of gains and losses relative to current wealth to explain why at all wealth levels individuals might both purchase insurance and take gambles.
- ▶ Kahneman and Tversky (1979) use this idea, together with probability weighting, to explain Allais and the other puzzles for expected-utility discussed before.
- ▶ Prospect theory takes the reference point as given. We will discuss some forward-looking models of endogenous reference point formation later.

Prospect Theory

Kahneman and Tversky (1979) describe a formula for evaluating simple prospects (gambles) of the form $(x, p; y, q)$ where $p + q \leq 1$. If $p + q < 1$, the remaining probability is understood to be assigned to zero.

Two ingredients in the model:

1. Value function $v(x)$, which satisfies $v(0) = 0$.
2. Weighting function $\pi(p)$, which satisfies $\pi(0) = 0$ and $\pi(1) = 1$.

Evaluation of lotteries (assume $x > y$):

$$V(x, 1) = V(x) = v(x)$$

$$V(x, p; y, q) = \pi(p)v(x) + \pi(q)v(y) \quad \text{if } p + q < 1 \text{ or } x \geq 0 \geq y$$

$$V(x, p; y, q) = v(y) + \pi(p)[v(x) - v(y)] \quad \text{if } p + q = 1 \text{ and } x > y > 0$$

Shape of the Value Function

Aversion to symmetric bets of the form $(x, 0.5; -x, 0.5)$ also implies

$$0 = v(0) > \pi(0.5)v(x) + \pi(0.5)v(-x) \Leftrightarrow v(x) < -v(-x).$$

- Significant concavity around the reference point 0 (e.g., kinks) can help explain scale-of-risk related paradoxes for EU (Rabin paradox, equity premium puzzle, etc.).

Experiment

KT (1979)

PROBLEM 13: Choose between

$$A : \begin{cases} 6,000 & \text{with probability } 0.25 \\ 0 & \text{with probability } 0.75 \end{cases} \quad \text{and} \quad C : \begin{cases} 4,000 & \text{with probability } 0.25 \\ 2,000 & \text{with probability } 0.25 \\ 0 & \text{with probability } 0.5 \end{cases}$$

$$\begin{aligned} B \succ A &\implies \pi(0.25)v(6000) < \pi(0.25)v(4000) + \pi(0.25)v(2000) \\ &\implies v(6000) < v(4000) + v(2000) \implies \text{concave for gains} \end{aligned}$$

PROBLEM 13': Choose between

$$C : \begin{cases} -6,000 & \text{with probability } 0.25 \\ 0 & \text{with probability } 0.75 \end{cases} \quad \text{and} \quad D : \begin{cases} -4,000 & \text{with probability } 0.25 \\ -2,000 & \text{with probability } 0.25 \\ 0 & \text{with probability } 0.5 \end{cases}$$

Recap: What Can Be Explained So Far

- ▶ We observed that significant concavity in the value function around the reference point might help to explain the scale of risk puzzles for expected utility:
 - ▶ Rabin paradox
 - ▶ Equity premium puzzle (see, Benartzi and Thaler (1995))
 - ▶ Kinks at 0 generate first-order risk aversion (as opposed to second-order risk aversion for expected utility)
- ▶ But we cannot yet explain the Allais paradox: Fixing the reference point at 0, the value function v is just an expected utility function and hence is subject to the paradox — unless we introduce probability weighting.

Probability Weighting

- ▶ Kahneman and Tversky (1979) show that explaining the Allais common consequence effect requires $\pi(0.66) + \pi(0.34) < 1$.
- ▶ They also argue that risk taking for low-probability implies overweighting of low probability events.

One problem for the KT weighting function is sensitivity to framing.

- ▶ For example suppose $2\pi(p) > \pi(2p)$ for some $p < 1/2$
- ▶ Then, for $x > 0$,

$$V(x, p; x, p; 0, 1 - 2p) = 2\pi(p)v(x) > \pi(2p)v(x) = V(x, 2p; 0, 1 - 2p).$$

- ▶ With examples like this in mind, KT proposed an **editing phase** where the framing of gambles is set: $(x, p; x, p; 0, 1 - 2p)$ must be treated like $(x, 2p; 0, 1 - 2p)$.

Problems with KT weighting Function

Unfortunately, imposing a framing requirement does not resolve the issue.

- ▶ Suppose that $2\pi(p) > \pi(2p)$, but alter the previous example slightly.
- ▶ Suppose $x > 0$. Then for $\epsilon > 0$ sufficiently small,

$$V(x, p; x-\epsilon, p; 0, 1-2p) = \pi(p)v(x) + \pi(p)v(x-\epsilon) > \pi(2p)v(x) = V(x, 2p; 0, 1-2p).$$

- ▶ This is a violation of stochastic dominance.
- ▶ In light of such examples, KT proposed that, in addition, stochastically dominated alternatives are dropped during the editing phase prior to applying the function V .

Rank-Dependent Utility

Definition

A **rank-dependent utility (RDU)** representation consists of :

1. a utility function $u : X \rightarrow \mathbb{R}$ for outcomes; and
2. a function $\varphi : [0, 1] \rightarrow [0, 1]$ (**probability distortion function**) that is continuous, nondecreasing, and onto.

The utility of a simple lottery of the form $P = (x_1, p_1; \dots; x_n, p_n)$ with outcomes ordered $x_1 < x_2 < \dots < x_n$ is

$$U(P) = \sum_{i=1}^n \pi_i u(x_i),$$

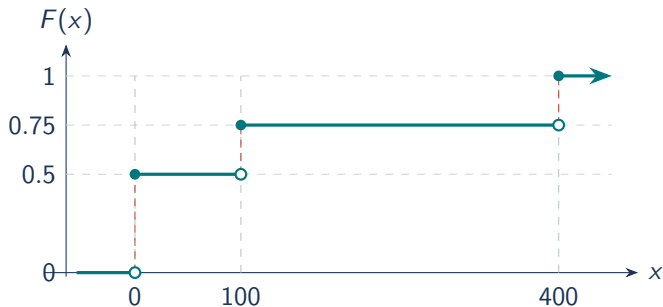
where

$$\pi_i = \varphi(p_1 + \dots + p_i) - \varphi(p_1 + \dots + p_{i-1})$$

for $i > 1$ and $\pi_1 = \varphi(p_1)$.

Illustration of Probability Distribution

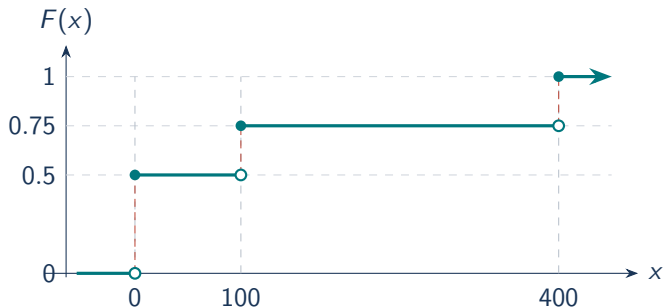
$$P = (0, 0.5; 100, 0.25; 400, 0.25), \quad F(x) = \Pr(X \leq x).$$



The cumulative distribution jumps by 0.5, 0.25, and 0.25 at 0, 100, and 400, respectively.

Illustration of EV

$$P = (0, 0.5; 100, 0.25; 400, 0.25)$$

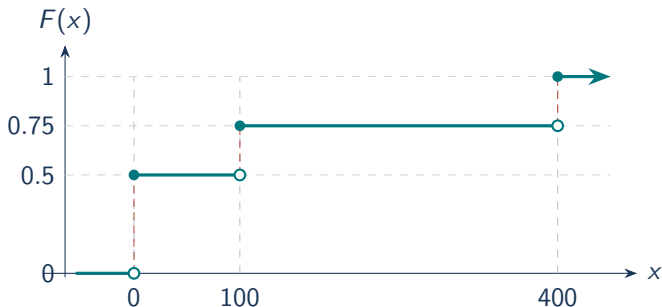


Each outcome receives the weight of the corresponding jump in F :

$$EV(P) = \int x dF(x) = 0.5 \times 0 + (0.75 - 0.5) \times 100 + (1 - 0.75) \times 400 = 125.$$

Illustration of EU

$$P = (0, 0.5; 100, 0.25; 400, 0.25)$$

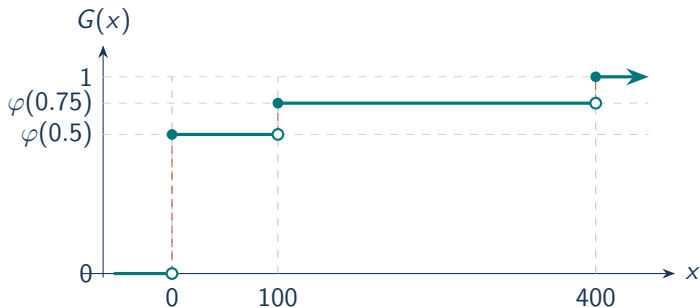


Expected utility uses the same jumps in F to weight utilities:

$$EU(P) = \int u(x) dF(x) = 0.5u(0) + (0.75 - 0.5)u(100) + (1 - 0.75)u(400).$$

Illustration of RDU

$$P = (0, 0.5; 100, 0.25; 400, 0.25), \quad G(x) = \varphi(F(x)).$$



Graph shown for $\varphi(t) = \sqrt{t}$.

Decision weights are the jumps in the transformed cumulative distribution:

$$\text{RDU}(P) = \varphi(0.5)u(0) + [\varphi(0.75) - \varphi(0.5)]u(100) + [1 - \varphi(0.75)]u(400).$$

Special Cases and Pessimism

- ▶ $\varphi(p) = p$: expected utility.
- ▶ $u(x) = x$: the dual theory of Yaari (1987).
- ▶ With the cumulative convention, $\varphi(p) \geq p$ means **pessimism**: more cumulative weight on low outcomes.

For a binary lottery with $x_1 < x_2$ and $u(x) = x$,

$$U(P) = \varphi(p_1)x_1 + [1 - \varphi(p_1)]x_2,$$
$$U(P) - \mathbb{E}[X] = [\varphi(p_1) - p_1](x_1 - x_2) \leq 0 \quad \text{if } \varphi(p_1) \geq p_1.$$

Two sources of risk attitudes

Utility curvature measures sensitivity to wealth. Probability distortion changes the weight assigned to different outcome ranks.

Weak and Strong Risk Aversion

Weak risk aversion: certainty versus a gamble

Prefer the objective mean for sure: $U(X) \leq u(\mathbb{E}[X])$. Increasing concave u and pessimism, $\varphi(p) \geq p$, are sufficient:

$$U(X) \leq u(\mathbb{E}_\varphi[X]) \leq u(\mathbb{E}[X]).$$

Strong risk aversion: one risky portfolio versus another

Reject every mean-preserving spread. Increasing concave u and increasing concave φ ensure this property and preserve second-order stochastic dominance.

Portfolio implication: Concavity of u alone does not justify eliminating all portfolios with extra mean-zero risk under RDU.

Rank-Dependent Utility for General lottery

Definition

For non-simple lotteries, **rank-dependent utility** simply transforms the cdf F using the distortion function φ :

$$U(F) = \int u(x) d(\varphi \circ F)(x).$$

For the cdf of a simple lottery, this reduces to the previous formula.

- ▶ The formula above is the cumulative version of RDU (transformation of the cumulative distribution).
 - ▶ Some papers instead use the “decumulative” formula for RDA (transformation of the decumulative distribution).
 - ▶ The cumulative and decumulative versions of RDU are equivalent after suitable transformation of the probability distortion function.

Decision Weights Follow the Portfolio

- ▶ p_s describes how likely state s is. Its decision weight π_s also depends on where the *total portfolio payoff* ranks.
- ▶ Example: a stock pays $S_L < S_H$, with probabilities p and $1 - p$.

Position	Payoff in the low-stock state	Weight of that state
Long stock	Lowest	$\varphi(p)$
Short stock	Highest	$1 - \varphi(1 - p)$

Implication for portfolio analysis

Evaluate the distribution of total wealth. Weights calculated for one asset or portfolio cannot generally be reused after a change in payoff ranks.

RDU and the Allais Paradox

Probability	0.01	0.10	0.89
a_1	€1M	€1M	€1M
a_2	0	€5M	€1M
a_3	€1M	€1M	0
a_4	0	€5M	0

Suppose $u(x) = x$.

$$\begin{aligned}a_1 \succ a_2 &\Leftrightarrow 1\delta_{1M} \succ 0.01\delta_0 + 0.89\delta_{1M} + 0.1\delta_{5M} \\&\Leftrightarrow u(1M) > u(0)\varphi(0.01) + u(1M)[\varphi(0.9) - \varphi(0.01)] + u(5M)[1 - \varphi(0.9)] \\&\Leftrightarrow \frac{1 - \varphi(0.9) + \varphi(0.01)}{1 - \varphi(0.9)} > \frac{5M}{1M}\end{aligned}$$

$$\begin{aligned}a_3 \prec a_4 &\Leftrightarrow 0.89\delta_0 + 0.11\delta_{1M} \prec 0.9\delta_0 + 0.1\delta_{5M} \\&\Leftrightarrow u(0)\varphi(0.89) + u(1M)[1 - \varphi(0.89)] < u(0)\varphi(0.9) + [1 - \varphi(0.9)]u(5M) \\&\Leftrightarrow \frac{1 - \varphi(0.89)}{1 - \varphi(0.9)} < \frac{5M}{1M}\end{aligned}$$

Exercises

Exercises

1. Demonstrate that RDU can resolve the common consequence paradox.
2. Demonstrate that RDU can resolve the common ratio paradox.

RDU and the Rabin Paradox

Take $u(x) = x$ for simplicity

$$\begin{aligned}U(w - L, 0.5; w + G, 0.5) &\geq U(W) \\ \Leftrightarrow \phi(0.5)(w - L) + (1 - \phi(0.5))(w + G) &\geq w \\ \Leftrightarrow \frac{G}{L} &\geq \frac{\phi(0.5)}{1 - \phi(0.5)}\end{aligned}$$

Example

Let $\phi(0.5) = 3/5$, then accept lottery iff $G/L \geq 3/2$.

L	G	accept /reject
100	125	reject
10 000	16 000	accept

Is that for the Rabin Paradox?

Most people do not consider the Rabin paradox resolved because of the **background risk**.

- ▶ Suppose I offer you a gain 110 or loss 100 lottery.
- ▶ This is probably not the only risk you face today, and certainly not the only risk you face over a longer period such as a year.
- ▶ If we integrate this background risk $\tilde{y}$ into your decision about the lottery $\tilde{x}$, then you are really choosing between $\tilde{x} + \tilde{y} + w$ and $\tilde{y} + w$ rather than $\tilde{x} + w$ and w .
- ▶ It turns out that sufficiently large $\tilde{y}$ makes it difficult to generate aversion to the small-stakes Rabin lotteries, so the paradox remains.
- ▶ One possible solution is narrow framing.

Cumulative Prospect Theory

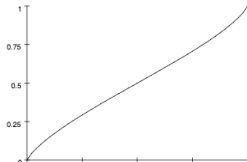
Tversky and Kahneman (1992) incorporate rank-dependent probability weighting into their model of prospect theory.

- ▶ They suggest decomposing any lottery into gains and losses and applying the rank-dependent formula to cumulative gain and loss probabilities separately.
- ▶ They suggest that a concave-convex shape for the weighting functions best fits the evidence (decumulative formula):

$$\varphi(p) = \frac{p^\beta}{(p^\beta + (1-p)^\beta)^{1/\beta}},$$

for some $\beta \in (0, 1)$.

- ▶ Overweights both low probability bad outcomes (certainty effect) and low probability good outcomes (possibility effect).



Loss Averse utility

- $u(x)$ is increasing in x , convex for loss, concave for gains, and first order concave at 0, that is

$$\lim_{x \rightarrow 0+} \frac{-u(-x)}{u(x)} = \lambda > 1.$$

- A useful parametrization

$$u(x) = \begin{cases} x^\beta & \text{for } x \geq 0 \\ -\lambda(-x)^\beta & \text{for } x \leq 0 \end{cases}$$

Exercise

Let $\beta = 1$. Characterize the range of β such that it can accommodate Rabin Paradox.

Second order risk aversion of EU

- ▶ Consider a lottery $(0.5, x + \sigma; 0.5, x - \sigma)$.
- ▶ Question: what risk premium π would you pay to avoid risk σ ?
- ▶ The risk premium r makes you be indifferent between

$$x \sim (0.5, x + \sigma + r(\sigma); 0.5, x - \sigma + r(\sigma)).$$

- ▶ Assume twice differentiable and take Taylor expansion.

$$\begin{aligned} u(x) &= u(x) + 1/2 u'(x) \cdot 2r(\sigma) + 1/4 u''(x) \cdot 2(\sigma^2 + r(\sigma)^2) + o(\sigma^2) \\ r(\sigma) &= -\frac{u''(x)}{u'(x)} \frac{\sigma^2 + r(\sigma)^2}{2} + o(\sigma^2) = \frac{\rho}{2} [\sigma^2 + r(\sigma)^2] + o(\sigma^2) \\ &\approx \frac{\rho}{2} \sigma^2 \end{aligned}$$

First-Order Risk Aversion from Probability Weighting

Let u be smooth with $u'(w) > 0$ and let $a = \varphi(1/2)$. A symmetric gamble pays $w - \sigma$ or $w + \sigma$, each with probability $1/2$. The compensation $r(\sigma)$ satisfies

$$u(w) = a u(w - \sigma + r) + (1 - a)u(w + \sigma + r).$$

A Taylor expansion gives

$$r(\sigma) = (2a - 1)\sigma + O(\sigma^2).$$

- ▶ If $a > 1/2$, the required compensation is first order in risk size, even with smooth utility. Under EU, $a = 1/2$ and this term vanishes.
- ▶ For a small long position θ with excess payoff $\mu \pm \sigma$,

$$\left. \frac{dU(w + \theta X)}{d\theta} \right|_{0+} = u'(w) [\mu - (2a - 1)\sigma].$$

A positive objective premium μ may be too small to induce participation.

RDU Portfolios in a Complete Market

Let $p_s > 0$ be state probabilities, $q_s > 0$ the prices of one unit of state-contingent wealth, and w initial wealth.

For a proposed ordering $x_1 \leq \dots \leq x_n$, set

$$P_s = \sum_{j=1}^s p_j, \quad \pi_s = \varphi(P_s) - \varphi(P_{s-1}), \quad P_0 = 0.$$

$$\max_{x_1, \dots, x_n} \sum_{s=1}^n \pi_s u(x_s) \quad \text{subject to} \quad \sum_{s=1}^n q_s x_s = w, \quad x_s \leq x_{s+1}.$$

- For the first-order conditions, assume strictly increasing, strictly concave, differentiable u and strictly increasing φ .
- The ordering constraints make the weights consistent with the payoffs. The full problem compares solutions across possible orderings.

Optimality Within a Payoff Ordering

$$\mathcal{L} = \sum_s \pi_s u(x_s) + \eta \left(w - \sum_s q_s x_s \right) + \sum_{s=1}^{n-1} \kappa_s (x_{s+1} - x_s).$$

For interior wealth levels, with $\kappa_0 = \kappa_n = 0$,

$$\begin{aligned}\pi_s u'(x_s) &= \eta q_s + \kappa_s - \kappa_{s-1}, \\ \kappa_s &\geq 0, \quad x_{s+1} - x_s \geq 0, \\ \kappa_s (x_{s+1} - x_s) &= 0.\end{aligned}$$

- ▶ When neighboring payoffs differ, the ordering constraint is slack.
- ▶ When it binds, consecutive states share the same payoff.
- ▶ Concavity makes these conditions sufficient within a fixed ordering. Comparing different orderings is still necessary for a global optimum.

Price per Unit of Decision Weight

Away from binding ordering constraints,

$$u'(x_s) = \eta \frac{q_s}{\pi_s}, \quad x_s = (u')^{-1} \left(\eta \frac{q_s}{\pi_s} \right).$$

	Expected utility	Rank-dependent utility
Relevant ratio	q_s/p_s	q_s/π_s
Probability input	Objective probability	Weight at the payoff's rank

- ▶ Lower q_s/π_s means a cheaper payoff per unit of decision weight, so the investor chooses more wealth in that state.
- ▶ q_s is a market price. π_s depends on the investor and the portfolio. It is not a fixed belief about state s .
- ▶ Overweighting a state raises the incentive to buy its payoff, holding the multiplier and ordering fixed. The budget couples all states.

Example: The Same Investor under EU and RDU

Take $w = 1$, $u(x) = x^{0.6}$ for $x \geq 0$, and a complete market:

State	s_1	s_2	s_3
Probability p_s	0.2	0.3	0.5
State price q_s	0.3	0.3	0.4
q_s/p_s	1.5	1	0.8

Under EU, $x_s \propto (p_s/q_s)^{2.5}$, with the budget fixing scale:

$$x^{\text{EU}} \approx (0.328, 0.903, 1.577).$$

Under RDU, use the cumulative distortion

$$\varphi(t) = \frac{t^{0.7}}{[t^{0.7} + (1-t)^{0.7}]^{1/0.7}}.$$

How does weighting change the *shape* of the optimal payoff?

Why Ignoring the Ordering Constraints Fails

With $u(x) = x^{0.6}$, the unconstrained formula is

$$x_s = \frac{(\pi_s/q_s)^{2.5}}{\sum_j q_j (\pi_j/q_j)^{2.5}}.$$

Assumed ordering	Weights (π_1, π_2, π_3)	Implied (x_1, x_2, x_3)
$x_1 < x_2 < x_3$	(0.2560, 0.2013, 0.5426)	(0.575, 0.315, 1.832)
$x_2 < x_1 < x_3$	(0.1293, 0.3281, 0.5426)	(0.096, 0.985, 1.689)

Both candidates reverse the assumed ordering of states 1 and 2.

Binding ordering constraints

Enforce the ordering constraints. In these two regions the constrained solution lies on the boundary $x_1 = x_2$. Then compare it with the solutions from the other four orderings.

Solution: Equal Payoffs in the Two Bad States

Set $x_1 = x_2 = a$ and $x_3 = b > a$. The combined bad outcome has probability 0.5, price 0.6, and weight $\varphi(0.5) = 0.457368$.

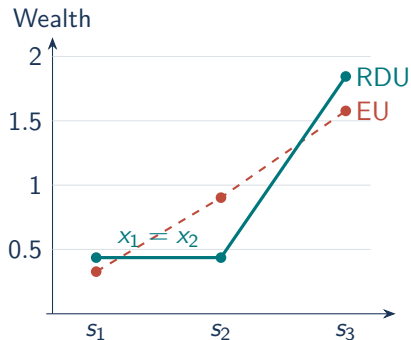
$$\max_{a \leq b} 0.457368 a^{0.6} + 0.542632 b^{0.6} \quad \text{subject to} \quad 0.6a + 0.4b = 1.$$

The first-order conditions give

$$\frac{b}{a} = \left(\frac{0.542632/0.4}{0.457368/0.6} \right)^{2.5} \approx 4.225, \quad \boxed{x^{\text{RDU}} \approx (0.437, 0.437, 1.845)}.$$

- ▶ Equal payoffs form a single outcome. Its weight is the *combined cumulative jump*, independent of the order within the tie.
- ▶ Checking all six orderings gives a maximum RDU value of 1.061815. The other distinct regional maxima are 1.040454 and 1.002129.

Portfolio Insurance and Upside Exposure



- ▶ Relative to EU, the investor raises wealth in the worst state and in the best state.
- ▶ Lower wealth in the middle state finances both changes.
- ▶ The flat lower segment resembles portfolio insurance. Upside exposure rises at the same time.

Tail weighting can generate downside protection and demand for large gains within the same portfolio. The precise shape depends on probabilities and prices.

Bunching and the General Solution Method

Suppose consecutive states $B = \{a, \dots, b\}$ share payoff z . Summing their first-order conditions cancels the internal multipliers:

$$u'(z) = \eta \frac{\sum_{s \in B} q_s}{\sum_{s \in B} \pi_s}, \quad \sum_{s \in B} \pi_s = \varphi(P_b) - \varphi(P_{a-1}),$$

when constraints at the block's outer boundaries are slack.

1. Propose an ordering and compute its cumulative decision weights.
2. Solve the concave problem with the ordering constraints. Pool adjacent states when unconstrained payoffs violate that ordering.
3. Compare the feasible solutions across orderings.

What survives from EU?

With equally probable states, an optimum assigns weakly higher payoffs to cheaper states. With unequal probabilities, inverse ordering by q_s/p_s need not hold. A consistent ordering alone does not prove global optimality.

Why Extra Mean-Zero Risk Can Be Attractive

Let $u(x) = x$ and use the pessimistic but nonconcave cumulative distortion

$$\varphi(t) = \begin{cases} 2t, & 0 \leq t \leq 1/4, \\ 3/8 + t/2, & 1/4 \leq t \leq 3/4, \\ t, & 3/4 \leq t \leq 1. \end{cases}$$

Lottery	Objective mean	RDU value
$X = (10, 0.4; 20, 0.6)$	16	14.250
$Y = (10, 0.4; 15, 0.3; 25, 0.3)$	16	14.875

Y adds a fair ± 5 risk when $X = 20$. Yet $Y \succ X$.

Portfolio implication

Preferring certainty to a fair gamble does not imply disliking every extra risk inside an existing portfolio. Diversification arguments require restrictions on probability weighting as well as on utility.

Ellsberg One Urn

Table: Ellsberg Example

	Red	Blue	Green
f	100	0	0
g	0	100	0
f'	100	0	100
g'	0	100	100

- ▶ 30 balls are Red and 60 balls are either Blue or Green.
- ▶ Do you prefer a bet on the known or unknown balls?

Ellsberg's Paradox (II)

- ▶ $f \succ g$ and $g' \succ f'$
- ▶ Under subjective expected utility, $f \succ g$ requires $p(R) > p(B)$, while $g' \succ f'$ requires $p(B) > p(R)$. No single prior supports both.
- ▶ DM often prefers known to unknown probabilities.
- ▶ This is known as “ambiguity aversion”.

Non-Bayesian Models of Ambiguity

Schmeidler (1989) emphasizes that a probability assessment alone does not describe how much information supports it.

- ▶ Choquet expected utility (CEU): beliefs take the form of a nonadditive capacity on events.
- ▶ Maxmin expected utility (MEU): evaluate an act using its worst expected utility over a set of plausible priors.
- ▶ Other approaches include variational preferences (Maccheroni, Marinacci and Rustichini, 2006) and smooth ambiguity (Klibanoff, Marinacci and Mukerji, 2005).

Risk versus ambiguity

Risk concerns outcomes under a specified probability distribution. Ambiguity concerns the probability model itself.

Maxmin Expected Utility

- ▶ X : outcomes; S : a finite state space.
- ▶ $\Delta(S)$: probability distributions on states.
- ▶ An act $f : S \rightarrow X$ specifies an outcome in each state.

Definition

A maxmin expected utility preference has representation

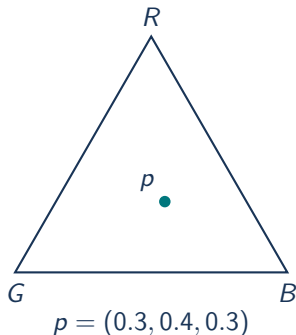
$$V(f) = \min_{p \in K} \sum_{s \in S} p(s) u(f(s)),$$

where $K \subseteq \Delta(S)$ is nonempty, compact and convex.

Utility $u : X \rightarrow \mathbb{R}$ need not be linear in monetary outcomes. In the lottery framework below, its extension to lotteries is affine in lottery probabilities.

The Probability Simplex

For $S = \{R, B, G\}$, a prior is $p = (p_R, p_B, p_G)$ with $p_R + p_B + p_G = 1$ and $p_s \geq 0$.

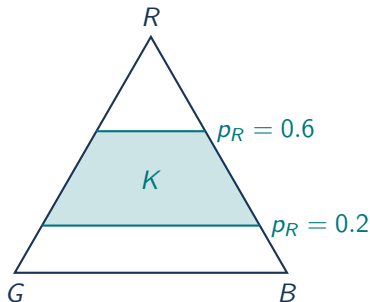


Vertices represent certainty about a state. This is a simplex of *beliefs*. The same geometry represents lotteries over prizes in the Marschak–Machina diagram.

A Set of Plausible Priors

Consider the set of priors

$$K = \{p \in \Delta(S) : 0.2 \leq p_R \leq 0.6\}.$$



A set of priors describes the range of plausible compositions. It does not assign a second-order probability to those priors.

Different Acts Can Have Different Worst-Case Priors

For $K = \{p \in \Delta(S) : 0.2 \leq p_R \leq 0.6\}$ and $u(x) = x$,

$$V(100\mathbf{1}_R) = 100 \min_{p \in K} p_R = 20,$$

$$V(100\mathbf{1}_{B \cup G}) = 100 \min_{p \in K} (1 - p_R) = 40.$$

- ▶ A bet on red has its worst case at $p_R = 0.2$.
- ▶ A bet against red has its worst case at $p_R = 0.6$.
- ▶ Together the two bets pay 100 for sure, so their combined value is 100.

Hedging ambiguity

The worst-case prior depends on the position. In general, $V(f + g)$ need not equal $V(f) + V(g)$, even with linear monetary utility.

Revisit Ellsberg One Urn

The known composition is $p_R = 1/3$, with the remaining $2/3$ split between blue and green. With $u(x) = x$, take

$$K = \{p \in \Delta(\{R, B, G\}) : p_R = 1/3\}.$$

For the four acts in the opening table,

$$V(f) = 100/3 > 0 = V(g), \quad V(g') = 200/3 > 100/3 = V(f').$$

Different bets expose the investor to different worst-case compositions.

Exercise

For two urns, one with a known 50–50 red/blue composition and one with unknown composition, construct a set of priors that explains a preference for betting on the known urn for either color.

The Anscombe–Aumann Framework

- ▶ X is the set of outcomes; $\Delta(X)$ is the set of simple lotteries.
- ▶ S is finite; $\Delta(S)$ is the probability simplex on states.
- ▶ $\mathcal{F} = \{f : S \rightarrow \Delta(X)\}$ is the set of acts.
- ▶ A lottery $\ell \in \Delta(X)$ also denotes the constant act $s \mapsto \ell$.
- ▶ Mixtures are statewise objective randomizations:

$$(\alpha f + (1 - \alpha)g)(s) = \alpha f(s) + (1 - \alpha)g(s).$$

Comparing $f(s)$ and $g(s)$ means comparing the corresponding constant acts.

Axioms: Order, Continuity and Monotonicity

Axiom 1: Weak order. $\succsim$ is complete and transitive.

Axiom 2: Continuity. If $f \succ g \succ h$, there exist $\alpha, \beta \in (0, 1)$ such that

$$\alpha f + (1 - \alpha)h \succ g \succ \beta f + (1 - \beta)h.$$

Axiom 3: Nondegeneracy. Some acts satisfy $f \succ g$.

Axiom 4: Monotonicity. If $f(s) \succsim g(s)$ for every $s \in S$, then $f \succsim g$.

Axioms: Independence and Ambiguity Aversion

Axiom 5: Certainty independence. For all $f, g \in \mathcal{F}$, constant lotteries $\ell \in \Delta(X)$ and $\alpha \in (0, 1)$,

$$f \succsim g \iff \alpha f + (1 - \alpha)\ell \succsim \alpha g + (1 - \alpha)\ell.$$

Axiom 5*: Full independence. Replace the constant ℓ in Axiom 5 by any act $h \in \mathcal{F}$.

Axiom 6: Ambiguity aversion. For $f, g \in \mathcal{F}$ and $\alpha \in [0, 1]$,

$$f \sim g \implies \alpha f + (1 - \alpha)g \succsim g.$$

Mixing indifferent acts can hedge ambiguity. Full independence would require indifference to every such mixture.

SEU and MEU Representation Theorems

In both representations, $u : \Delta(X) \rightarrow \mathbb{R}$ is nonconstant and affine.

Subjective expected utility. Axioms 1–4 and 5* hold if and only if some $p \in \Delta(S)$ satisfies

$$f \succsim g \iff \sum_s p(s)u(f(s)) \geq \sum_s p(s)u(g(s)).$$

Maxmin expected utility. Axioms 1–6 hold if and only if some nonempty compact convex $K \subseteq \Delta(S)$ satisfies

$$f \succsim g \iff \min_{p \in K} \sum_s p(s)u(f(s)) \geq \min_{p \in K} \sum_s p(s)u(g(s)).$$

Utility is unique up to a positive affine transformation. The SEU prior and the MEU closed convex set of priors are unique.

Proof of MEU: Utility and Constant Equivalents

1. Restrict preferences to constant lotteries. Certainty independence becomes ordinary mixture independence, so the VNM theorem gives a nonconstant affine utility u .
2. For each act f , choose its best and worst state lotteries ℓ_H, ℓ_L . Monotonicity gives $\ell_H \succsim f \succsim \ell_L$.
3. Continuity gives a constant lottery ℓ_f with $f \sim \ell_f$. Define $V(f) = u(\ell_f)$.

$$f \succsim g \iff V(f) \geq V(g).$$

The utility value of ℓ_f is unique, though the lottery itself need not be. If $\ell_H \sim \ell_L$, any mixture of them has that same value.

Proof of MEU: The Utility Functional

Normalize u so that its range $J = u(\Delta(X))$ contains $[-1, 1]$. Since S is finite, define

$$\mathcal{A} = J^S, \quad I(u \circ f) = V(f).$$

- ▶ I is well-defined: equal utility vectors give statewise indifference, hence indifference between acts by monotonicity.
- ▶ I evaluates constant vectors at their constant value: $I(c\mathbf{1}) = c$ for $c \in J$.
- ▶ If $a \geq b$ statewise, then $I(a) \geq I(b)$.

The domain $\mathcal{A}$ need not be closed under addition or arbitrary scaling. We first establish identities where their arguments lie in the domain.

Proof of MEU: Positive Homogeneity and Extension

Choose a constant lottery ℓ_0 with $u(\ell_0) = 0$. If $f \sim \ell_f$ and $0 < t < 1$, certainty independence implies

$$tf + (1 - t)\ell_0 \sim t\ell_f + (1 - t)\ell_0, \quad I(ta) = tI(a).$$

The same identity holds for $t > 1$ whenever $a, ta \in \mathcal{A}$, by applying the previous identity with $1/t$.

Extension to all utility vectors

For $z \in \mathbb{R}^S$, choose $t > 0$ small enough that $tz \in [-1, 1]^S$. Define $\hat{I}(z) = I(tz)/t$. Homogeneity makes this independent of t .

This is the unique positively homogeneous extension. It is monotone and satisfies $\hat{I}(c\mathbf{1}) = c$ for every real c .

Proof of MEU: Translation Invariance

Certainty independence and constant equivalents imply, for $a \in \mathcal{A}$ and $c \in J$,

$$I\left(\frac{a + c\mathbf{1}}{2}\right) = \frac{I(a) + c}{2}.$$

For arbitrary $z \in \mathbb{R}^S$ and $c \in \mathbb{R}$, choose $t > 0$ so that $tz \in \mathcal{A}$ and $tc \in J$. Then

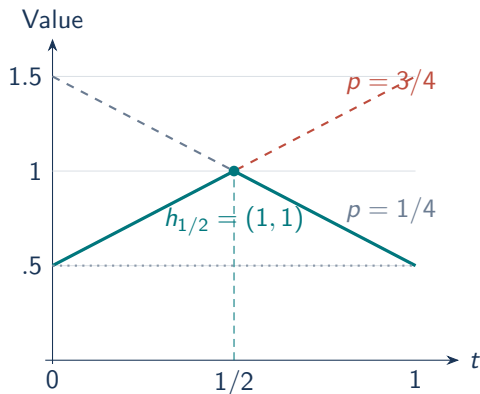
$$\begin{aligned}\widehat{I}(z + c\mathbf{1}) &= \frac{2}{t} I\left(\frac{tz + tc\mathbf{1}}{2}\right) \\ &= \frac{I(tz)}{t} + c = \widehat{I}(z) + c.\end{aligned}$$

Monotonicity now also gives continuity:

$$|\widehat{I}(z) - \widehat{I}(y)| \leq \|z - y\|_{\infty}.$$

MEU Geometry: Mixing Can Hedge Ambiguity

In utility units, let $f = (0, 2)$, $g = (2, 0)$ and $K = \{(p, 1 - p) : 1/4 \leq p \leq 3/4\}$. For $h_t = (1 - t)f + tg$, the utility vector is $(2t, 2 - 2t)$.



Each prior gives a straight line:

$$\mathbb{E}_{1/4}[u(h_t)] = 1.5 - t,$$

$$\mathbb{E}_{3/4}[u(h_t)] = 0.5 + t.$$

MEU takes the lower line.

$$V(h_t) = \min\{1.5 - t, 0.5 + t\}.$$

The endpoints both have value 0.5.
Their equal mixture pays utility 1 in either state.

The lower envelope is concave: the mixture can lie above the chord joining two indifferent acts. This illustrates the ambiguity-aversion step of the proof.

Proof of MEU: Superadditivity

For $a, b \in \mathbb{R}^S$, put $c = \hat{l}(a) - \hat{l}(b)$ and $b' = b + c\mathbf{1}$. Translation invariance gives $\hat{l}(b') = \hat{l}(a)$.

Scale a, b' into $\mathcal{A}$. Ambiguity aversion for the associated indifferent acts, followed by homogeneity, gives

$$\hat{l}\left(\frac{a + b'}{2}\right) \geq \hat{l}(a).$$

Using translation invariance again,

$$\hat{l}\left(\frac{a + b}{2}\right) \geq \hat{l}(a) - \frac{c}{2} = \frac{\hat{l}(a) + \hat{l}(b)}{2}.$$

Thus $\hat{l}(a + b) \geq \hat{l}(a) + \hat{l}(b)$. Positive homogeneity and superadditivity imply concavity.

Proof of MEU: Supporting Probabilities

A finite continuous concave function on $\mathbb{R}^S$ has a supporting affine function at every a .
Thus some vector p_a satisfies

$$\hat{l}(z) \leq \hat{l}(a) + p_a \cdot (z - a) \quad \text{for all } z.$$

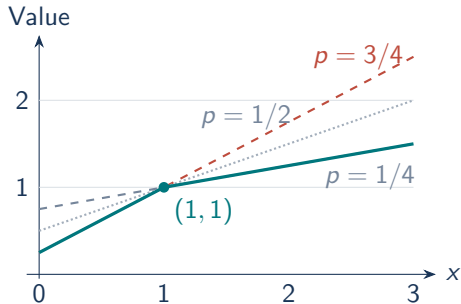
- ▶ Set $z = 0$ and $z = 2a$. Homogeneity gives $p_a \cdot a = \hat{l}(a)$, hence $\hat{l}(z) \leq p_a \cdot z$.
- ▶ For a state indicator e_s , monotonicity gives $0 \leq \hat{l}(e_s) \leq p_a(s)$.
- ▶ For constant vectors, $c = \hat{l}(c\mathbf{1}) \leq c \sum_s p_a(s)$ for both signs of c , so $\sum_s p_a(s) = 1$.

Every supporting vector p_a is therefore a probability in $\Delta(S)$.

MEU Geometry: Recovering the Supporting Priors

Take a slice of utility space: $z = (x, 1)$. In the same example,

$$\hat{l}(x, 1) = \min\left\{\frac{3}{4}x + \frac{1}{4}, \frac{1}{4}x + \frac{3}{4}\right\}.$$



1. Concavity: supporting affine functions lie above and touch $\hat{l}$.
2. Homogeneity: supports are linear in the full vector z .
3. Monotonicity and constants: the coefficients satisfy $p_s \geq 0$ and $\sum_s p_s = 1$.

Every support is an expectation $p \cdot z$; at each z , one attains equality:

$$\hat{l}(z) = \min_{p \in K} p \cdot z.$$

Proof of MEU: The Set of Priors

Define

$$K = \{p \in \Delta(S) : p \cdot z \geq \hat{l}(z) \text{ for every } z \in \mathbb{R}^S\}.$$

- ▶ K is nonempty by the supporting-probability argument.
- ▶ It is closed and convex, hence compact as a subset of $\Delta(S)$.
- ▶ Every $p \in K$ satisfies $p \cdot a \geq \hat{l}(a)$. A supporting probability $p_a \in K$ attains equality at a .

Therefore

$$\hat{l}(a) = \min_{p \in K} p \cdot a, \quad V(f) = \min_{p \in K} \sum_s p(s) u(f(s)).$$

Conversely, this representation satisfies Axioms 1–6: minima of affine functions are concave and preserve constant mixtures.

Capacity as a Belief

Definition

A capacity is a set function $\mu : 2^S \rightarrow [0, 1]$ such that

$$\mu(\emptyset) = 0, \quad \mu(S) = 1, \quad A \subseteq B \implies \mu(A) \leq \mu(B).$$

Definition

A capacity is convex (supermodular) if

$$\mu(A \cup B) + \mu(A \cap B) \geq \mu(A) + \mu(B) \quad \text{for every } A, B \subseteq S.$$

Its core is

$$C(\mu) = \{p \in \Delta(S) : p(A) \geq \mu(A) \text{ for every } A \subseteq S\}.$$

For a convex capacity on finite S , the core is nonempty and the Choquet integral equals minimum expected value over this core.

The Choquet Integral

For a real-valued function $z : S \rightarrow \mathbb{R}$, define

$$\int_S^C z \, d\mu = \int_{-\infty}^0 [\mu(\{s : z(s) > t\}) - 1] \, dt + \int_0^{\infty} \mu(\{s : z(s) > t\}) \, dt.$$

On finite S , order states so that $z(s_1) \leq \dots \leq z(s_n)$. With $A_i = \{s_i, \dots, s_n\}$ and $A_{n+1} = \emptyset$,

$$\int_S^C z \, d\mu = \sum_{i=1}^n [\mu(A_i) - \mu(A_{i+1})] z(s_i).$$

Equivalently,

$$\int_S^C z \, d\mu = z(s_1) + \sum_{i=2}^n [z(s_i) - z(s_{i-1})] \mu(A_i).$$

These formulas apply to negative as well as positive payoffs.

Choquet Evaluation of the Ellsberg Bets

Let $u(x) = x$, $\mu(\emptyset) = 0$, $\mu(S) = 1$, and set

$$\begin{aligned}\mu(R) &= 1/3, & \mu(B) &= \mu(G) = 1/4, \\ \mu(B \cup G) &= 2/3, & \mu(R \cup G) &= \mu(R \cup B) = 1/2.\end{aligned}$$

This is a monotone capacity. A bet paying 100 on event A has value $100\mu(A)$, so

$$V(f) = 100/3 > 25 = V(g), \quad V(g') = 200/3 > 50 = V(f').$$

It explains the Ellsberg choices. This particular capacity is not convex:

$$\mu(R \cup B) = 1/2 < 1/3 + 1/4 = \mu(R) + \mu(B).$$

Thus fitting these choices alone does not establish the full ambiguity-aversion axiom.

Comonotonicity

Two real-valued functions z, y are comonotonic when

$$(z(s) - z(t))(y(s) - y(t)) \geq 0 \quad \text{for every } s, t \in S.$$

Two acts are comonotonic if they never rank two states in opposite directions according to the preference over their state lotteries. Equivalently, their utility vectors are comonotonic.

Axiom 7: Comonotonic independence. For pairwise comonotonic $f, g, h \in \mathcal{F}$ and $\alpha \in (0, 1)$,

$$f \succsim g \iff \alpha f + (1 - \alpha)h \succsim \alpha g + (1 - \alpha)h.$$

Mixing comonotonic acts does not hedge opposing state rankings.

Choquet Expected Utility

Theorem (Schmeidler)

A preference on $\mathcal{F}$ satisfies Axioms 1–4 and 7 if and only if there exist a capacity μ on S and a nonconstant affine $u : \Delta(X) \rightarrow \mathbb{R}$ such that

$$f \succsim g \iff \int_S^C u(f(s)) \, d\mu \geq \int_S^C u(g(s)) \, d\mu.$$

The capacity is unique, and u is unique up to a positive affine transformation.

If μ is convex, CEU has the maxmin representation

$$V(f) = \min_{p \in C(\mu)} \sum_s p(s) u(f(s)).$$

General CEU does not require a convex capacity; general MEU need not be CEU.

Revisit Ellsberg Two Urns

Urn 1 has a known 50–50 composition; urn 2 has an unknown red/blue split. In each state, the first letter records urn 1's draw, the second urn 2's draw.

	RR	RB	BR	BB
f_1 : red from urn 1	100	100	0	0
f_2 : blue from urn 1	0	0	100	100
f_3 : red from urn 2	100	0	100	0
f_4 : blue from urn 2	0	100	0	100

The ambiguity-averse choices are $f_1 \succ f_3$ and $f_2 \succ f_4$.

Exercise

Construct a capacity explaining both choices. One route is to take the lower envelope of independent priors with $p_1(R) = 1/2$ and $p_2(R) \in [0, 1]$.

Ambiguity and a No-Trade Interval

An asset pays $X(L) = 1$, $X(H) = 3$. Let $u(x) = x$ and $\mu(L) = 0.3$, $\mu(H) = 0.4$.

$$V(X) = 1 + (3 - 1)\mu(H) = 1.8.$$

The short position pays -3 in H and -1 in L :

$$V(-X) = -3 + [-1 - (-3)]\mu(L) = -2.4.$$

At price p , with the risk-free gross return normalized to 1,

$$V(X - p) = 1.8 - p, \quad V(p - X) = p - 2.4.$$

- For $1.8 < p < 2.4$, zero strictly dominates every nonzero position.
- At $p = 1.8$, the investor is indifferent between zero and long positions. At $p = 2.4$, zero and short positions are indifferent.

Thus zero is optimal on the closed interval $[1.8, 2.4]$, with strict preference for zero only in its interior.

Portfolio Analysis under Risk

Money has price and terminal payoff 1. A risky asset with price p has terminal payoff $\tilde{v} \sim N(\hat{v}, \hat{\sigma}^2)$, with $\hat{\sigma}^2 > 0$.

- ▶ CARA utility: $u(W) = -\exp(-W)$.
- ▶ Budget: $w = m + px$. Terminal wealth: $\widetilde{W} = m + x\tilde{v} = w + x(\tilde{v} - p)$.
- ▶ Normality gives the certainty equivalent

$$\text{CE}(x) = w + (\hat{v} - p)x - \frac{1}{2}\hat{\sigma}^2 x^2.$$

With unrestricted borrowing and short selling,

$$x_R^*(p) = \frac{\hat{v} - p}{\hat{\sigma}^2}.$$

Here $\hat{\sigma}$ is the standard deviation; $\hat{\sigma}^2$ is the variance.

Portfolio Analysis under Ambiguity

Retain the same budget and CARA utility. The investor considers all models

$$\tilde{v} \sim N(\bar{v}, \sigma^2), \quad \bar{v} \in [v_{\min}, v_{\max}], \quad \sigma \in [\sigma_{\min}, \sigma_{\max}], \quad 0 < \sigma_{\min} \leq \sigma_{\max}.$$

Maxmin expected utility is equivalent to maximizing the worst-case certainty equivalent:

$$\max_x \min_{\bar{v}, \sigma} \left\{ w + (\bar{v} - p)x - \frac{1}{2}\sigma^2 x^2 \right\}.$$

- ▶ For any nonzero position, the worst-case variance is $\sigma_{\max}^2$.
- ▶ A long position uses the lowest mean $v_{\min}$.
- ▶ A short position uses the highest mean $v_{\max}$.

The worst-case model changes with the sign of the position.

Optimal Demand under Ambiguity

Maximizing the worst-case certainty equivalent gives

$$x_A^*(p) = \begin{cases} (v_{\min} - p)/\sigma_{\max}^2, & p < v_{\min}, \\ 0, & v_{\min} \leq p \leq v_{\max}, \\ (v_{\max} - p)/\sigma_{\max}^2, & p > v_{\max}. \end{cases}$$

- ▶ Ambiguity about the mean creates a no-participation interval.
- ▶ Higher worst-case variance reduces the magnitude of a nonzero position.
- ▶ Even at the interval endpoints, zero is the unique optimum here because the variance penalty is strictly positive for every nonzero position.

This last property differs from the preceding example with linear utility.

Portfolio Analysis: Equilibrium

Normalize investor mass to 1. All investors have the same CARA utility, unrestricted borrowing and short selling, and access to risk-free money.

- ▶ A fraction $1 - \lambda$ uses $N(\hat{v}, \hat{\sigma}^2)$.
- ▶ A fraction λ uses the ambiguous set of normal models.
- ▶ Assume $0 \leq \lambda < 1$, $\hat{\sigma}^2 > 0$ and $\sigma_{\max}^2 > 0$.
- ▶ The risky asset has aggregate supply $\bar{x}$ per investor.

Market clearing requires

$$(1 - \lambda)x_R^*(p) + \lambda x_A^*(p) = \bar{x}.$$

Aggregate demand is continuous, strictly decreasing, and ranges from $+\infty$ to $-\infty$. Hence the equilibrium price is unique.

Equilibrium Price and Participation

Define the price when ambiguity-averse investors hold zero,

$$\hat{p} = \hat{v} - \frac{\hat{\sigma}^2 \bar{x}}{1 - \lambda},$$

and, for $b \in \{v_{\min}, v_{\max}\}$,

$$p_b = \frac{\lambda \hat{\sigma}^2 b + (1 - \lambda) \sigma_{\max}^2 \hat{v} - \bar{x} \sigma_{\max}^2 \hat{\sigma}^2}{\lambda \hat{\sigma}^2 + (1 - \lambda) \sigma_{\max}^2}.$$

Condition	Equilibrium price	Ambiguity-averse demand
$\hat{p} < v_{\min}$	$p_{v_{\min}}$	$x_A^* > 0$
$v_{\min} \leq \hat{p} \leq v_{\max}$	$\hat{p}$	$x_A^* = 0$
$\hat{p} > v_{\max}$	$p_{v_{\max}}$	$x_A^* < 0$

If $\bar{x} > 0$ and $\hat{v} \in [v_{\min}, v_{\max}]$, only the first two regimes arise. In the participating regime both types then hold positive positions.